MTH636 – Game Theory

Lecture 26: Subgames and Subgame Perfect Equilibrium

1 Subgames of an Extensive Form Game

Recall that a general extensive form game is denoted

$$\Gamma = \langle N,\, V,\, E,\, n_0,\, (V_i)_{i\in N\cup\{0\}},\, \mathcal{H},\, \big(A(h)\big)_{h\in\bigcup_i H_i},\, (p_n)_{n\in V_0},\, (u_i)_{i\in N}\rangle.$$

Definition: Subgame

For a game $\Gamma = \langle N,\, V,\, E,\, n_0,\, (V_i)_{i\in N\cup\{0\}},\, \mathcal{H},\, (A(h))_{h\in\bigcup_i H_i},\, (p_n)_{n\in V_0},\, (u_i)_{i\in N}\rangle$, a **subgame** starting at a node $y \in V$ is itself a game

$$\Gamma_y = \langle N,\, V_y,\, E_y,\, y,\, (V_i)^y_{i\in N\cup\{0\}},\, \mathcal{H}^y,\, \big(A(h)\big)^y_{h\in\bigcup_i H_i},\, (p_n)_{n\in V_0^y},\, (u_i)_{i\in N}\rangle,$$

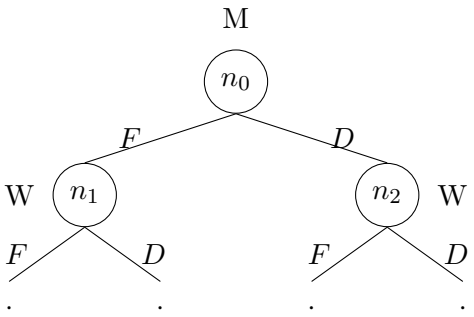
where V_y is the set of all nodes reachable from y, E_y is the restriction of edges to V_y, and every information set h must either be **entirely contained** in V_y or entirely in $V \setminus V_y$.

The condition on information sets is crucial: a subgame may only begin at a node y if no information set is "split" by y – that is, no information set contains both a node inside V_y and a node outside V_y.

1.1 Illustrative Examples of Subgames

Example 1 – Battle of the Sexes (extensive form)

Consider the sequential Battle of the Sexes game. The Man (player M) moves first at the root n_0 and can choose F (Football) or D (Dance). The Woman (player W) then moves at the resulting node.



Here the information set of the Woman (player W) is $\{n_1, n_2\}$, shown with a dashed arc connecting the two nodes. Because both n_1 and n_2 belong to the *same* information set, neither n_1 alone nor n_2 alone can serve as the root of a subgame – the information set

would be split. The *only* proper subgame is the entire game itself (i.e. the game is its own unique subgame).

Example 2 – Three-player game with information sets

Consider a game with three players I, II, III whose tree has nodes $n_0, n_1, \ldots, n_6$.

- Player I moves at n_0 (actions A and B).
- Player II has an information set $\{n_1, n_2\}$ and moves with actions ℓ and r (from n_1) and L, R (from n_2).
- Player III has an information set $\{n_3, n_4, n_5, n_6\}$ and takes actions a, b at each node.

Because player III's information set $\{n_3, n_4, n_5, n_6\}$ spans nodes reached via *both* branches of player I's move, no subgame can be started at any node $n_3, \ldots, n_6$ individually. The entire game is therefore its own only subgame.

2 Subgame Perfect Equilibrium (SPE)

Definition: Subgame Perfect Equilibrium (SPE)

A strategy profile σ is a **Subgame Perfect Equilibrium (SPE)** if it is a Nash Equilibrium in *all* subgames of the game.

SPE refines Nash Equilibrium by eliminating non-credible threats: a player cannot credibly threaten to take a suboptimal action in a subgame that would never be reached in equilibrium, because the SPE requirement forces rational play even in off-path subgames.

3 Examples of Finding SPE

Example A

Consider a two-player sequential game with the following structure. Player I moves first at n_0 choosing L or R. After L the game reaches a subgame at n_2 where player II chooses ℓ or r; the payoff from L alone is $(1, -1)$. After R the game reaches a subgame at n_3 where player II again chooses ℓ or r.

The terminal payoffs are:

	ℓ	r
L	$(1, -1)$	$(-1, 1)$
R	$(-1, 1)$	$(1, -2)$

Nash Equilibria found: (L, ℓ) and $(-1, 1)$ – two NE exist.

SPE: (L, ℓ) – only one SPE.

Observe the game tree with player I at n_0 branching into L (to n_2, player II) and R (to n_3, player II), where player II's information set $\{n_2, n_3\}$ is circled. The four terminal payoffs are $(1, -1), (-1, 1), (-1, 1), (1, -2)$ from left to right.

Note

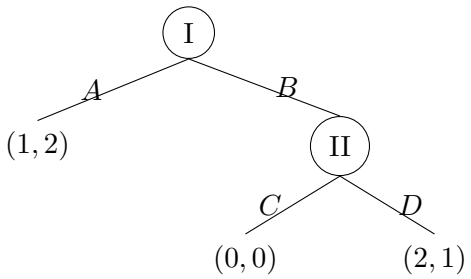
There are two Nash Equilibria (L, ℓ) and $(-1, 1)$, but *only one* SPE: (L, ℓ).

Example B – Two-stage sequential game

Player I moves at n_0: actions A or B.

- If A: payoff $(1, 2)$ immediately.
- If B: play reaches player II at node u_1, who chooses C or D, giving payoffs $(0, 0)$ or $(2, 1)$ respectively.

Game tree:



The normal form matrix (Player I rows, Player II columns):

	C	D
A	$(1,\ 2)$	$(1,\ 2)$
B	$(0,\ 0)$	$(2,\ 1)$

Two NE: (A, C) and (B, D).

One SPE: (B, D).

The strategy (A, C) is an NE: if Player I plays A, Player II's choice is irrelevant (payoff is $(1, 2)$ regardless). However, if the game ever reached Player II's node, Player II would prefer D (payoff 1 to II) over C (payoff 0). The threat of C is non-credible. The SPE is therefore (B, D).

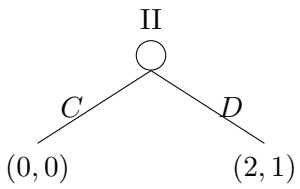
Example C – Mixed strategy NE and SPE

Consider the same game structure as Example B but now allowing mixed strategies. The NE of the game are:

$$(A,\ \alpha[C] + (1 - \alpha)[D]), \quad \alpha \geq \tfrac{1}{2}$$

and (B, D).

The SPE are those NE that also constitute an NE in the subgame rooted at player II's node. In the subgame at n_1:



In the subgame Γ_{n_1}, the strategy $\alpha[C]+(1-\alpha)[D]$ is *not* an NE if $\alpha > 0$: Player II strictly prefers D (payoff 1) over C (payoff 0). Therefore only D is an NE of the subgame.

- The mixed strategy $(A,\ \alpha[C]+(1-\alpha)[D])$ with $1 > \alpha \geq \frac{1}{2}$: **SPE** (Player I plays A, so subgame is off-path; but Player II must play D in the subgame – contradiction with $\alpha > 0$). *Not* an SPE for $\alpha > 0$.
- (B, D): **SPE**.

Note

$(\alpha[C],\ (1-\alpha)[D])$ is not an NE of the subgame at n_1 for any $\alpha > 0$. The only NE of that subgame is D. Therefore the unique SPE is (B, D).

Example D – Larger sequential game

Player I at n_0 chooses A or B.

- $A \to$ payoff $(1, 2)$.
- $B \to$ Player II at node with children n_3 (from C) and n_5 (from D).
  - $C \to$ payoff $(0, 0)$.
  - $D \to$ payoff $(1, 1)$.

Normal form:

	C	D
A	$(1,\ 2)$	$(1,\ 2)$
B	$(0,\ 0)$	$(1,\ 1)$

NE of the game: (A, C) (not SPE), (A, D) (SPE), and $\left(\frac{1}{2}[A] + \frac{1}{2}[B],\ D\right)$ (SPE).

(B, D) is also an SPE.

Key Takeaway

The SPE concept eliminates Nash Equilibria that rely on non-credible threats. Whenever a player's strategy prescribes an action in a subgame that is irrational for that player (given the subgame), that NE is *not* an SPE.